

GRADE 7 ACCELERATED · UNIT 4

Solving Problems with Rational Numbers

Six lessons in teaching order · the next slide jumps to each one

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Adding and Subtracting in Scientific Notation

MA.8.NSO.1.5

I can add and subtract two numbers written in scientific notation, and write the answer in scientific notation.

Last unit you learned to write numbers in scientific notation.

Today you add and subtract them — the powers must match.

Warm-Up

On your own. Four minutes. No notes.

1. Write 0.00062 in scientific notation.
2. How many times larger is 8×10^7 than 2×10^4 ?
3. Find the value: $\sqrt{144}$
4. Find the sum: $0.49 + 1.2$

Warm-Up

Answers.

1. Write 0.00062 in scientific notation. **6.2×10^{-4}**
2. How many times larger is 8×10^7 than 2×10^4 ? **4,000**
3. Find the value: $\sqrt{144}$ **12**
4. Find the sum: $0.49 + 1.2$ **1.69**

Notes

Copy the table.

I. A power of ten is a place value

10^3	10^4	10^5	10^6
thousands	ten thousands	hundred thousands	millions

$$49,500 + 121,000 = 170,500$$

$$4.95 \times 10^4 + 12.1 \times 10^4 = 17.05 \times 10^4 = 1.705 \times 10^5$$

Two numbers can be added only when they carry the same power of ten.

Notes

II. Same power: add or subtract the coefficients

$$(5.2 \times 10^6) + (3.1 \times 10^6) = 8.3 \times 10^6$$

$$(7 \times 10^{-8}) + (6 \times 10^{-8}) = 13 \times 10^{-8} = 1.3 \times 10^{-7}$$

Add or subtract the coefficients. The power of ten does not change.

Then check the coefficient: if it is not between 1 and 10, the answer is not finished.

Notes

III. Different powers: rewrite one of them first

$$(9.4 \times 10^5) - (8.2 \times 10^3)$$

$$= (9.4 \times 10^5) - (0.082 \times 10^5) = 9.318 \times 10^5$$

$$(1.3 \times 10^3) + (3.4 \times 10^5) = 3.413 \times 10^5, \quad \text{never} \quad 4.7 \times 10^8$$

Rewrite one number so both carry the same power of ten, then add or subtract the coefficients.

Exponent up one, decimal point one place left. The value does not move.

Example 1

Same power, and the answer is not finished.

$$(8.6 \times 10^7) + (5.9 \times 10^7)$$

Add. Write the answer in scientific notation.

Example 1

Worked.

$$(8.6 \times 10^7) + (5.9 \times 10^7) = 14.5 \times 10^7 \quad \text{same power: add the coefficients}$$

$$14.5 \times 10^7 = 1.45 \times 10^8 \quad \text{14.5 is not less than 10: point one place left, exponent up one}$$

$$145,000,000 \quad \text{check in standard form}$$

Answer: 1.45×10^8

Your Turn

Same steps, your numbers. Boards up when done.

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

Your Turn

Answer.

$$(4.8 \times 10^{-3}) + (7.5 \times 10^{-3})$$

12.3×10^{-3} , then repair

Answer: 1.23×10^{-2}

Example 2

Different powers, and a subtraction.

$$(7.3 \times 10^6) - (4.5 \times 10^5)$$

Subtract. Write the answer in scientific notation.

Example 2

Worked.

$$4.5 \times 10^5 = 0.45 \times 10^6 \quad \text{rewrite over } 10^6$$

$$(7.3 \times 10^6) - (0.45 \times 10^6) = 6.85 \times 10^6 \quad \text{same power: subtract the coefficients}$$

$$7,300,000 - 450,000 = 6,850,000 \quad \text{check in standard form}$$

Answer: 6.85×10^6

Your Turn

Same steps, your numbers. Boards up when done.

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

Your Turn

Answer.

$$(5.6 \times 10^{-4}) - (9.0 \times 10^{-6})$$

0.09×10^{-4} comes off 5.6×10^{-4}

Answer: 5.51×10^{-4}

Whiteboards • Question 1 of 9

Boards up on three.

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

Answer it.

Same power.

Whiteboards • Question 1 of 9

Answer.

$$(3.2 \times 10^5) + (4.1 \times 10^5)$$

add the coefficients

Answer: 7.3×10^5

Whiteboards • Question 2 of 9

Boards up on three.

A stadium sold 4.2×10^4 tickets on Friday
and 3.5×10^4 on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A stadium sold 4.2×10^4 tickets on Friday
and 3.5×10^4 on Saturday. Each ticket cost 38 dollars.

How many tickets were sold on the two days?

$$4.2 + 3.5$$

Answer: 7.7×10^4

Whiteboards • Question 3 of 9

Boards up on three.

$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

Answer it.

Finish the answer.

Whiteboards • Question 3 of 9

Answer.

$$(6 \times 10^{-5}) + (7 \times 10^{-5})$$

13×10^{-5} is not finished

Answer: 1.3×10^{-4}

Whiteboards • Question 4 of 9

Boards up on three.

$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

Answer it.

Finish the answer.

Whiteboards • Question 4 of 9

Answer.

$$(5.6 \times 10^8) - (4.9 \times 10^8)$$

0.7×10^8 is not finished either

Answer: 7×10^7

Whiteboards • Question 5 of 9

Boards up on three.

$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

A. 4.7×10^5

B. 4.7×10^8

C. 3.413×10^5

D. 3.53×10^5

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$(1.3 \times 10^3) + (3.4 \times 10^5)$$

Which is the sum?

A. 4.7×10^5

B. 4.7×10^8

C. 3.413×10^5

D. 3.53×10^5

Answer: C — 3.413×10^5

Whiteboards • Question 6 of 9

Boards up on three.

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

A. 12.3×10^6

B. 12.3×10^{12}

C. 1.23×10^6

D. 1.23×10^7

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(2.5 \times 10^6) + (9.8 \times 10^6)$$

Which is the sum, written in scientific notation?

A. 12.3×10^6

B. 12.3×10^{12}

C. 1.23×10^6

D. 1.23×10^7

Answer: D — 1.23×10^7

Whiteboards • Question 7 of 9

Boards up on three.

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

Answer it.

Rewrite one of them first.

Whiteboards • Question 7 of 9

Answer.

$$(7.8 \times 10^{-2}) - (6.4 \times 10^{-3})$$

$$6.4 \times 10^{-3} = 0.64 \times 10^{-2}$$

$$\textbf{Answer: } 7.16 \times 10^{-2}$$

Whiteboards • Question 8 of 9

Boards up on three.

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of n ?

Answer it.

Rewrite the answer over 10^6 first.

Whiteboards • Question 8 of 9

Answer.

$$(4.5 \times 10^6) + (n \times 10^6) = 1.02 \times 10^7$$

What is the value of n ?

$$1.02 \times 10^7 = 10.2 \times 10^6$$

Answer: $n = 5.7$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Devon writes $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$.

Name the two mistakes he made, and give the correct sum.

Write your answer in sentences.

Two mistakes, then the sum.

Whiteboards • Question 9 of 9

Answer.

Devon writes $(1.3 \times 10^3) + (3.4 \times 10^5) = 4.7 \times 10^8$.

Name the two mistakes he made, and give the correct sum.

one mistake in the coefficients, one in the exponents

Answer: 3.413×10^5

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Add and subtract numbers written in scientific notation — HUR

Multiplying and Dividing in Scientific Notation

MA.8.NSO.1.5

I can multiply and divide two numbers written in scientific notation, and write the answer in scientific notation.

Yesterday the powers of ten had to match before you could add.

Today the powers of ten do not have to match at all.

Warm-Up

On your own. Four minutes. No notes.

1. $(3.4 \times 10^6) + (2.8 \times 10^6)$
2. Write as one power: $\frac{10^9}{10^4}$
3. Find the value: $\sqrt{169}$
4. Find the quotient: $\frac{4.8}{1.6}$

Warm-Up

Answers.

1. $(3.4 \times 10^6) + (2.8 \times 10^6)$ **6.2×10^6**

2. Write as one power: $\frac{10^9}{10^4}$ **10^5**

3. Find the value: $\sqrt{169}$ **13**

4. Find the quotient: $\frac{4.8}{1.6}$ **3**

Notes

Copy both lines.

I. Multiplying: regroup, then use the laws

$$(2 \times 10^4)(3 \times 10^5) = (2 \cdot 3)(10^4 \cdot 10^5) = 6 \times 10^9$$

$$6 \times 10^9, \quad \text{never} \quad 6 \times 10^{20}$$

Multiply the coefficients. Add the exponents — that is the product of powers law, not a new rule.

Notes

II. Dividing: the same move with the quotient law

$$\frac{8.4 \times 10^9}{2.1 \times 10^4} = \frac{8.4}{2.1} \times 10^{9-4} = 4 \times 10^5$$

$$\frac{3 \times 10^8}{3 \times 10^{-4}} = 1 \times 10^{8-(-4)} = 1 \times 10^{12}$$

Divide the coefficients. Subtract the exponents, top minus bottom.

Notes

III. Finish the answer — both directions

$$(8 \times 10^2)(7.5 \times 10^4) = 60 \times 10^6 = 6 \times 10^7$$

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}} = 0.53 \times 10^{-2} = 5.3 \times 10^{-3}$$

Every answer ends with a coefficient at least 1 and less than 10.

Example 1

Multiplying, and the answer needs finishing.

$$(4.5 \times 10^{-3})(6 \times 10^7)$$

Multiply. Write the answer in scientific notation.

Example 1

Worked.

$$(4.5 \times 10^{-3})(6 \times 10^7) = (4.5 \cdot 6)(10^{-3} \cdot 10^7) \quad \text{regroup}$$

$$= 27 \times 10^4 \quad \text{multiply the coefficients, add the exponents}$$

$$27 \times 10^4 = 2.7 \times 10^5 \quad \text{27 is not less than 10: point one place left, exponent up one}$$

Answer: 2.7×10^5

Your Turn

Same steps, your numbers. Boards up when done.

$$(2.5 \times 10^6)(8 \times 10^{-2})$$

Your Turn

Answer.

$$(2.5 \times 10^6)(8 \times 10^{-2})$$

20×10^4 , then repair

Answer: 2×10^5

Example 2

Multiplication and division in one expression.

$$\frac{(8 \times 10^2)(7.5 \times 10^4)}{5 \times 10^2}$$

Evaluate. Write the answer in scientific notation.

Example 2

Worked.

$$(8 \times 10^2)(7.5 \times 10^4) = 60 \times 10^6 \quad \text{the top, regrouped}$$

$$\frac{60 \times 10^6}{5 \times 10^2} = 12 \times 10^4 \quad \text{divide the coefficients, subtract the exponents}$$

$$12 \times 10^4 = 1.2 \times 10^5 \quad \text{finish}$$

Answer: 1.2×10^5

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{(6 \times 10^5)(4 \times 10^{-3})}{8 \times 10^4}$$

Your Turn

Answer.

$$\frac{(6 \times 10^5)(4 \times 10^{-3})}{8 \times 10^4}$$

$$24 \times 10^2 \text{ over } 8 \times 10^4$$

Answer: 3×10^{-2}

Whiteboards • Question 1 of 9

Boards up on three.

$$(3 \times 10^4)(2 \times 10^5)$$

Answer it.

Multiply the fronts, add the backs.

Whiteboards • Question 1 of 9

Answer.

$$(3 \times 10^4)(2 \times 10^5)$$

$$3 \cdot 2 \text{ and } 10^4 \cdot 10^5$$

$$\textbf{Answer: } 6 \times 10^9$$

Whiteboards • Question 2 of 9

Boards up on three.

A printer makes 4×10^3 copies per hour.

It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet.

How many copies does it make?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A printer makes 4×10^3 copies per hour.

It runs for 2.5×10^2 hours. Paper costs 3 cents a sheet.

How many copies does it make?

10×10^5 , then finish

Answer: 1×10^6

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{9.6 \times 10^8}{1.2 \times 10^3}$$

Answer it.

Top minus bottom.

Whiteboards • Question 3 of 9

Answer.

$$\frac{9.6 \times 10^8}{1.2 \times 10^3}$$

$$9.6 \div 1.2 \text{ and } 10^8 \div 10^3$$

Answer: 8×10^5

Whiteboards • Question 4 of 9

Boards up on three.

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}}$$

Answer it.

Finish the answer.

Whiteboards • Question 4 of 9

Answer.

$$\frac{2.65 \times 10^{11}}{5 \times 10^{13}}$$

0.53×10^{-2} is not finished

Answer: 5.3×10^{-3}

Whiteboards • Question 5 of 9

Boards up on three.

$$(2 \times 10^4)(3 \times 10^5)$$

Which is the product?

A. 6×10^{-1}

B. 6×10^{20}

C. 6×10^9

D. 5×10^9

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$(2 \times 10^4)(3 \times 10^5)$$

Which is the product?

A. 6×10^{-1}

B. 6×10^{20}

C. 6×10^9

D. 5×10^9

Answer: C — 6×10^9

Whiteboards • Question 6 of 9

Boards up on three.

$$(5 \times 10^6)(4 \times 10^3)$$

Which is the product, written in scientific notation?

A. 2×10^9

B. 20×10^9

C. 20×10^{18}

D. 2×10^{10}

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(5 \times 10^6)(4 \times 10^3)$$

Which is the product, written in scientific notation?

A. 2×10^9

B. 20×10^9

C. 20×10^{18}

D. 2×10^{10}

Answer: D — 2×10^{10}

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{3 \times 10^8}{3 \times 10^{-4}}$$

Answer it.

Top minus bottom, and the bottom is negative.

Whiteboards • Question 7 of 9

Answer.

$$\frac{3 \times 10^8}{3 \times 10^{-4}}$$

$$8 - (-4) = 12$$

Answer: 1×10^{12}

Whiteboards • Question 8 of 9

Boards up on three.

$$\frac{8.4 \times 10^7}{n \times 10^2} = 4.2 \times 10^5$$

What is the value of n ?

Answer it.

Look at the coefficients alone.

Whiteboards • Question 8 of 9

Answer.

$$\frac{8.4 \times 10^7}{n \times 10^2} = 4.2 \times 10^5$$

What is the value of n ?

$$8.4 \div n = 4.2$$

Answer: $n = 2$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

For $\frac{7.2 \times 10^9}{2.4 \times 10^4}$ Marcus writes 4.8×10^5

and Sara writes 3×10^{13}

Name what each one did wrong, and give the correct quotient.

Write your answer in sentences.

Each one got exactly one half right.

Whiteboards • Question 9 of 9

Answer.

For $\frac{7.2 \times 10^9}{2.4 \times 10^4}$ Marcus writes 4.8×10^5

and Sara writes 3×10^{13}

Name what each one did wrong, and give the correct quotient.

one wrong in front, one wrong behind

Answer: 3×10^5

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Multiply numbers written in scientific notation — YZU

Divide numbers written in scientific notation — SGT

Real-World Problems and Significant Digits

MA.8.NSO.1.6

I can solve a real-world problem with numbers in scientific notation and report the answer with the right number of significant digits.

Yesterday you multiplied and divided in scientific notation.

Today the numbers are measurements, each with limits.

Warm-Up

On your own. Four minutes. No notes.

1. $(4 \times 10^5)(3 \times 10^{-2})$
2. $(6.3 \times 10^7) - (4.1 \times 10^6)$
3. Write 0.00902 in scientific notation.
4. Round 4.738 to the nearest hundredth.

Warm-Up

Answers.

1. $(4 \times 10^5)(3 \times 10^{-2})$ **1.2×10^4**
2. $(6.3 \times 10^7) - (4.1 \times 10^6)$ **5.89×10^7**
3. Write 0.00902 in scientific notation. **9.02×10^{-3}**
4. Round 4.738 to the nearest hundredth. **4.74**

Notes

Copy the table.

I. Which digits are significant

Measurement	Digits	Why
4,207	4	the middle zero counts
0.0058	2	leading zeros never count
3.200	4	trailing zeros count after a decimal point
52,000	2	no decimal point, so they do not

Notes

II. Multiplying or dividing: count the digits

$$(2.5 \times 10^2)(4.13 \times 10^2) = 1.0325 \times 10^5$$

2 significant digits and 3, so the answer keeps 2: 1.0×10^5

Multiply or divide: the answer takes the smaller of the two significant-digit counts.

Notes

Copy all three lines.

III. Adding or subtracting: a different rule

$$1,234 + 5.6 = 1,239.6$$

1,234 stops at the ones place; 5.6 stops at the tenths

so the sum stops at the ones place: $1,240 = 1.240 \times 10^3$

Add or subtract: the answer stops at the leftmost of the two last-digit places.

Example 1

A product of two measurements.

A solar array measures 1.25×10^2 m by 8.6×10^1 m.

What is its area?

Give the answer to the correct number of significant digits.

Example 1

Worked.

$$(1.25 \times 10^2)(8.6 \times 10^1) = 10,750 \quad \text{multiply in full}$$

$1.25 \rightarrow 3$ digits, $8.6 \rightarrow 2$ digits *the answer keeps 2*

$$10,750 \rightarrow 11,000 = 1.1 \times 10^4 \quad \text{round once, at the end}$$

Answer: 1.1×10^4 square meters

Your Turn

Same steps, your numbers. Boards up when done.

A field measures 4.6×10^2 m by 2.35×10^2 m.

What is its area?

Your Turn

Answer.

A field measures 4.6×10^2 m by 2.35×10^2 m.

What is its area?

108,100, kept to two significant digits

Answer: 1.1×10^5 square meters

Example 2

A difference of two measurements.

A container holds 3.09×10^{-3} cubic feet of liquid.

3.472×10^{-5} cubic feet evaporates.

How much is left? Give the answer with the correct precision.

Example 2

Worked.

$$0.00309 - 0.00003472 = 0.00305528 \quad \text{subtract in full}$$

$$0.00309 \rightarrow \text{stops at } 10^{-5} \quad \text{the leftmost last-digit place}$$

$$0.00305528 \rightarrow 0.00306 = 3.06 \times 10^{-3} \quad \text{round there}$$

Answer: 3.06×10^{-3} cubic feet

Your Turn

Same steps, your numbers. Boards up when done.

A road is 3.85×10^3 m long.

A section 2.74×10^2 m long is repaved.

How much of the road is not repaved?

Your Turn

Answer.

A road is 3.85×10^3 m long.

A section 2.74×10^2 m long is repaved.

How much of the road is not repaved?

3,576, stopped at the tens place

Answer: 3.58×10^3 m

Whiteboards • Question 1 of 9

Boards up on three.

How many significant digits are in 0.00470 ?

Answer it.

Leading zeros, then trailing zeros.

Whiteboards • Question 1 of 9

Answer.

How many significant digits are in 0.00470 ?

4, 7 and the last 0

Answer: 3

Whiteboards • Question 2 of 9

Boards up on three.

A pond's surface measures 1.4×10^2 m by 8.5×10^1 m.

The pond is 3 meters deep.

Find the surface area. Round to the correct number of significant digits.

Answer it.

Whiteboards • Question 2 of 9

Answer.

A pond's surface measures 1.4×10^2 m by 8.5×10^1 m.

The pond is 3 meters deep.

Find the surface area. Round to the correct number of significant digits.

11,900, kept to two digits

Answer: 1.2×10^4 square meters

Whiteboards • Question 3 of 9

Boards up on three.

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5 .

How should the answer be written?

Answer it.

Which count wins?

Whiteboards • Question 3 of 9

Answer.

Two measurements are multiplied. One has 3 significant digits and the other has 5. The exact product is 7.4126×10^5 .

How should the answer be written?

the smaller count

Answer: 7.41×10^5

Whiteboards • Question 4 of 9

Boards up on three.

$(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured.

Write the sum with the correct precision.

Answer it.

Where does each one's last digit sit?

Whiteboards • Question 4 of 9

Answer.

$(2.5 \times 10^3) + (4.71 \times 10^2)$ meters, both measured.

Write the sum with the correct precision.

2,500 stops at the hundreds

Answer: 3.0×10^3 m

Whiteboards • Question 5 of 9

Boards up on three.

How many significant digits are in 0.03080 ?

A. 4

B. 2

C. 5

D. 3

Answer it.

Whiteboards • Question 5 of 9

Answer.

How many significant digits are in 0.03080 ?

A. 4

B. 2

C. 5

D. 3

Answer: A — 4

Whiteboards • Question 6 of 9

Boards up on three.

A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm.

Which is the area, to the correct significant digits?

A. $26.1 \times 10^4 \text{ cm}^2$

B. $2.6 \times 10^5 \text{ cm}^2$

C. $2.6 \times 10^4 \text{ cm}^2$

D. $2.61 \times 10^5 \text{ cm}^2$

Answer it.

Whiteboards • Question 6 of 9

Answer.

A rectangle measures 6.0×10^2 cm by 4.35×10^2 cm.

Which is the area, to the correct significant digits?

A. $26.1 \times 10^4 \text{ cm}^2$

B. $2.6 \times 10^5 \text{ cm}^2$

C. $2.6 \times 10^4 \text{ cm}^2$

D. $2.61 \times 10^5 \text{ cm}^2$

Answer: B — $2.6 \times 10^5 \text{ cm}^2$

Whiteboards • Question 7 of 9

Boards up on three.

A car travels 4.20×10^2 km in 3.0 hours.

What is its average speed, to the right number of significant digits?

Answer it.

Count both measurements.

Whiteboards · Question 7 of 9

Answer.

A car travels 4.20×10^2 km in 3.0 hours.

What is its average speed, to the right number of significant digits?

140 exactly, kept to two digits

Answer: 1.4×10^2 km/h

Whiteboards • Question 8 of 9

Boards up on three.

A measured length times a measured width gives an area of $3.6 \times 10^3 \text{ m}^2$. The length measured $9.0 \times 10^1 \text{ m}$.

What did the width measure, to two significant digits?

Answer it.

Divide, then count.

Whiteboards • Question 8 of 9

Answer.

A measured length times a measured width gives an area of $3.6 \times 10^3 \text{ m}^2$. The length measured $9.0 \times 10^1 \text{ m}$.

What did the width measure, to two significant digits?

$$3,600 \div 90$$

Answer: $4.0 \times 10^1 \text{ m}$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Ellis says: 1,234 has 4 significant digits and 5.6 has 2,
so the sum $1,234 + 5.6$ must be written with 2: 1.2×10^3 .

Explain why he is wrong, and give the correct sum.

Write your answer in sentences.

Which rule is for which operation?

Whiteboards • Question 9 of 9

Answer.

Ellis says: 1,234 has 4 significant digits and 5.6 has 2,
so the sum $1,234 + 5.6$ must be written with 2: 1.2×10^3 .

Explain why he is wrong, and give the correct sum.

products count digits; sums count places

Answer: $1,240 = 1.240 \times 10^3$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Add, subtract, multiply, and divide numbers written in scientific notation: word problems — D5S

Evaluating Expressions with Radicals

MA.8.NSO.1.7

I can evaluate an expression that contains square roots, cube roots and integer exponents.

Yesterday every number was a measurement.

Today the radical returns — its bar is a grouping symbol.

Warm-Up

On your own. Four minutes. No notes.

1. How many significant digits are in 0.0250 ?
2. $(3 \times 10^4)(2 \times 10^{-7})$
3. Find the value: $\sqrt{196}$
4. Evaluate: $20 - 3 \cdot 4$

Warm-Up

Answers.

1. How many significant digits are in 0.0250 ? **3**
2. $(3 \times 10^4)(2 \times 10^{-7})$ **6×10^{-3}**
3. Find the value: $\sqrt{196}$ **14**
4. Evaluate: $20 - 3 \cdot 4$ **8**

Notes

Copy all three lines.

I. The radical bar is a grouping symbol

$$\sqrt{25} - \sqrt{36} + \sqrt{1} = 5 - 6 + 1 = 0$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4$$

$$\sqrt{9 + 16} = \sqrt{25} = 5, \quad \text{never} \quad 3 + 4$$

Everything under the bar is evaluated before the root is taken.

Notes

II. Square roots and cube roots are not the same shape

$$\sqrt{49} = 7 \qquad \sqrt[3]{64} = 4 \qquad \sqrt[3]{-64} = -4$$

$$\sqrt{\frac{1}{9}} = \frac{1}{3} \qquad \sqrt[3]{\frac{1}{125}} = \frac{1}{5}$$

An odd root keeps the sign of its radicand. A square root sign never gives a negative value.

Notes

III. Radicals alongside exponents

$$5^{-3} = \frac{1}{5^3} = \frac{1}{125} \qquad \sqrt{4^3 - 39} = \sqrt{25} = 5$$

$$\sqrt{4^3 - 39} \cdot 5^{-3} = 5 \cdot \frac{1}{125} = \frac{1}{25}$$

Finish each piece — the radical, the power — and only then put them together.

Example 1

A power and a radical in one expression.

$$\left(-\frac{1}{2}\right)^2 + \sqrt{2^3 + 8}$$

Evaluate.

Example 1

Worked.

$$\left(-\frac{1}{2}\right)^2 = \frac{1}{4} \quad \text{square the whole base, sign included}$$

$$\sqrt{2^3 + 8} = \sqrt{16} = 4 \quad \text{finish under the bar, then take the root}$$

$$\frac{1}{4} + 4 = \frac{17}{4} \quad \text{combine}$$

Answer: 17/4

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

Your Turn

Answer.

$$\left(-\frac{1}{3}\right)^2 - \sqrt[3]{2^2 + 4}$$

1/9 minus 2

Answer: $-17/9$

Example 2

A radical inside a product.

$$\sqrt[3]{27} - 1.4(\sqrt{3^2 - 5})$$

Evaluate.

Example 2

Worked.

$$\sqrt[3]{27} = 3, \quad 3^2 - 5 = 4 \quad \text{the two innermost pieces}$$

$$\sqrt{4} = 2, \quad 1.4(2) = 2.8 \quad \text{the root, then the product}$$

$$3 - 2.8 = 0.2 \quad \text{subtract last}$$

Answer: 0.2

Your Turn

Same steps, your numbers. Boards up when done.

$$\sqrt[3]{-8} + 2.5(\sqrt{5^2 - 9})$$

Your Turn

Answer.

$$\sqrt[3]{-8} + 2.5(\sqrt{5^2 - 9})$$

−2 plus 10

Answer: 8

Whiteboards • Question 1 of 9

Boards up on three.

$$\sqrt{121}$$

Answer it.

What times itself?

Whiteboards • Question 1 of 9

Answer.

$$\sqrt{121}$$

$$11 \cdot 11$$

Answer: 11

Whiteboards • Question 2 of 9

Boards up on three.

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A square patio has an area of 169 square feet.

Fencing for it costs 12 dollars a foot.

How long is one side of the patio?

$$\sqrt{169}$$

Answer: 13 feet

Whiteboards • Question 3 of 9

Boards up on three.

$$\sqrt[3]{-125}$$

Answer it.

What number, three times?

Whiteboards • Question 3 of 9

Answer.

$$\sqrt[3]{-125}$$

$$(-5)(-5)(-5)$$

Answer: -5

Whiteboards • Question 4 of 9

Boards up on three.

$$\sqrt{2^3 + 17}$$

Answer it.

Finish under the bar first.

Whiteboards • Question 4 of 9

Answer.

$$\sqrt{2^3 + 17}$$

$$8 + 17 = 25$$

Answer: 5

Whiteboards • Question 5 of 9

Boards up on three.

$$\sqrt{9 + 16}$$

Which is the value?

A. 7

B. 25

C. 12.5

D. 5

Answer it.

Whiteboards · Question 5 of 9

Answer.

$$\sqrt{9 + 16}$$

Which is the value?

A. 7

B. 25

C. 12.5

D. 5

Answer: D — 5

Whiteboards • Question 6 of 9

Boards up on three.

$$\sqrt[3]{-64}$$

Which is the value?

A. -4

B. not a real number

C. -8

D. 4

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\sqrt[3]{-64}$$

Which is the value?

A. -4

B. not a real number

C. -8

D. 4

Answer: A — -4

Whiteboards • Question 7 of 9

Boards up on three.

$$\sqrt{\frac{1}{9}} + \sqrt[3]{\frac{1}{27}}$$

Answer it.

Top and bottom each get the root.

Whiteboards • Question 7 of 9

Answer.

$$\sqrt{\frac{1}{9}} + \sqrt[3]{\frac{1}{27}}$$

$\frac{1}{3}$ and $\frac{1}{3}$

Answer: $\frac{2}{3}$

Whiteboards • Question 8 of 9

Boards up on three.

$$\sqrt{4^3 - 39} \cdot 5^{-3}$$

Answer it.

Finish each piece, then multiply.

Whiteboards • Question 8 of 9

Answer.

$$\sqrt{4^3 - 39} \cdot 5^{-3}$$

5 times one hundred twenty-fifth

Answer: 1/25

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Simon says that $\sqrt[3]{-27}$ is not a real number,
because you cannot take the root of a negative.
Explain why he is wrong, and give the value.

Write your answer in sentences.

Which root is he thinking of?

Whiteboards • Question 9 of 9

Answer.

Simon says that $\sqrt[3]{-27}$ is not a real number,

because you cannot take the root of a negative.

Explain why he is wrong, and give the value.

a cube root is not a square root

Answer: -3

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. This skill continues tomorrow: one assignment, due at the start of the class after 4.06.

Today's skills — all required

Evaluate radical expressions — D8J

Order of Operations with Radicals

MA.8.NSO.1.7

I can evaluate a multi-step expression with rational numbers, exponents and radicals by reading its structure.

Yesterday the radical bar grouped whatever sat under it.

Today: all the grouping symbols — structure sets the order.

Warm-Up

On your own. Four minutes. No notes.

1. $\sqrt{5^2 - 9}$
2. How many significant digits are in 1.020 ?
3. Find the value: $\left(\frac{2}{3}\right)^{-2}$
4. Evaluate: $12 \div 3 \cdot 2$

Warm-Up

Answers.

1. $\sqrt{5^2 - 9}$ **4**

2. How many significant digits are in 1.020 ? **4**

3. Find the value: $\left(\frac{2}{3}\right)^{-2}$ **9/4**

4. Evaluate: $12 \div 3 \cdot 2$ **8**

Notes

Copy the table.

I. The structure, not a mnemonic

	Level	What it means
1	Grouping symbols	(), [], the radical bar, the fraction bar
2	Powers and roots	the same level as each other
3	Multiply and divide	left to right
4	Add and subtract	left to right

Innermost grouping symbol first, then out.

Notes

II. Left to right means left to right

$$12 \div 3 \cdot 2 = 4 \cdot 2 = 8, \quad \text{never} \quad 12 \div 6 = 2$$

$$10 - 4 + 3 = 6 + 3 = 9, \quad \text{never} \quad 10 - 7 = 3$$

Multiplication and division are one level. Addition and subtraction are one level. Left to right, both times.

Notes

III. The fraction bar groups, top and bottom

$$\frac{2^3 - 5}{6^2 - 27} = \frac{3}{9} = \frac{1}{3}$$

$$\frac{2^3 - 5}{6^2} - \sqrt{\frac{4}{9} - \frac{1}{3}} = \frac{1}{12} - \frac{1}{3} = -\frac{1}{4}$$

Finish the numerator and finish the denominator before you divide.

Example 1

A grouping symbol inside a grouping symbol.

$$4(8.63 - (3.2)^2) + \sqrt[3]{125}$$

Evaluate.

Example 1

Worked.

$$(3.2)^2 = 10.24 \quad \text{innermost: the power inside the parentheses}$$

$$8.63 - 10.24 = -1.61, \quad 4(-1.61) = -6.44 \quad \begin{array}{l} \text{finish the} \\ \text{parentheses,} \\ \text{then multiply} \end{array}$$

$$-6.44 + 5 = -1.44 \quad \text{the cube root is 5; add last}$$

Answer: -1.44

Your Turn

Same steps, your numbers. Boards up when done.

$$3(5.4 - (2.1)^2) + \sqrt{49}$$

Your Turn

Answer.

$$3(5.4 - (2.1)^2) + \sqrt{49}$$

$$3(0.99) + 7$$

Answer: 9.97

Example 2

A fraction bar and a radical in one expression.

$$\frac{2^3 - 5}{6^2} - \sqrt{\frac{4}{9} - \frac{1}{3}}$$

Evaluate.

Example 2

Worked.

$$\frac{2^3 - 5}{6^2} = \frac{3}{36} = \frac{1}{12} \quad \text{numerator and denominator, each finished}$$

$$\frac{4}{9} - \frac{1}{3} = \frac{1}{9}, \quad \sqrt{\frac{1}{9}} = \frac{1}{3} \quad \text{under the bar, then the root}$$

$$\frac{1}{12} - \frac{1}{3} = -\frac{1}{4} \quad \text{subtract last}$$

Answer: $-1/4$

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{3^2 - 1}{4^2} + \sqrt{\frac{7}{16} - \frac{3}{16}}$$

Your Turn

Answer.

$$\frac{3^2 - 1}{4^2} + \sqrt{\frac{7}{16} - \frac{3}{16}}$$

a half plus a half

Answer: 1

Whiteboards • Question 1 of 9

Boards up on three.

$$12 \div 3 \cdot 2$$

Answer it.

Left to right.

Whiteboards · Question 1 of 9

Answer.

$$12 \div 3 \cdot 2$$

$12 \div 3$ first

Answer: 8

Whiteboards • Question 2 of 9

Boards up on three.

A square tile has side $\sqrt{196}$ centimeters.

Its diagonal is about 19.8 centimeters.

What is the perimeter of the tile?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A square tile has side $\sqrt{196}$ centimeters.

Its diagonal is about 19.8 centimeters.

What is the perimeter of the tile?

4 sides of 14

Answer: 56 centimeters

Whiteboards • Question 3 of 9

Boards up on three.

$$10 - 4 + 3$$

Answer it.

Left to right.

Whiteboards • Question 3 of 9

Answer.

$$10 - 4 + 3$$

10 - 4 first

Answer: 9

Whiteboards • Question 4 of 9

Boards up on three.

$$\frac{2^3 - 5}{6^2 - 27}$$

Answer it.

Finish the top and the bottom.

Whiteboards • Question 4 of 9

Answer.

$$\frac{2^3 - 5}{6^2 - 27}$$

3 over 9

Answer: 1/3

Whiteboards • Question 5 of 9

Boards up on three.

$$2 + 3 \cdot 4^2$$

Which is the value?

A. 146

B. 50

C. 26

D. 80

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$2 + 3 \cdot 4^2$$

Which is the value?

A. 146

B. 50

C. 26

D. 80

Answer: B — 50

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{6^2 - 4}{2^3}$$

Which is the value?

A. 4

B. 35.5

C. 0.5

D. 256

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{6^2 - 4}{2^3}$$

Which is the value?

A. 4

B. 35.5

C. 0.5

D. 256

Answer: A — 4

Whiteboards • Question 7 of 9

Boards up on three.

$$-2^2 + (-2)^2$$

Answer it.

What is each exponent attached to?

Whiteboards · Question 7 of 9

Answer.

$$-2^2 + (-2)^2$$

-4 and +4

Answer: 0

Whiteboards • Question 8 of 9

Boards up on three.

$$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$$

Answer it.

Inside first.

Whiteboards • Question 8 of 9

Answer.

$$12\left(\left(\frac{1}{2}\right)^3 + 3^{-2}\right)$$

one eighth plus one ninth

Answer: 17/6

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Roscoe says that $(\sqrt{n} - 11)^2$ is always positive,
because a square is always positive.
Is he right? Give a value of n that settles it.

Write your answer in sentences.

Try to make the inside zero.

Whiteboards • Question 9 of 9

Answer.

Roscoe says that $(\sqrt{n} - 11)^2$ is always positive,
because a square is always positive.

Is he right? Give a value of n that settles it.

never negative is not the same as always positive

Answer: No — at $n = 121$ the value is 0

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate radical expressions — D8J

Real-World Order of Operations

MA.8.NSO.1.7

I can substitute values into a formula and evaluate it, reading what the situation does rather than hunting for key words.

Yesterday you read an expression's structure.

Today the expression is in a sentence — not in key words.

Warm-Up

On your own. Four minutes. No notes.

1. $-5^2 + (-5)^2$
2. $\sqrt{144} + 2^{-1}$
3. Find the value: $\left(\frac{3}{4}\right)^{-1}$
4. Evaluate $2x^2$ when $x = 3$.

Warm-Up

Answers.

1. $-5^2 + (-5)^2$ **0**

2. $\sqrt{144} + 2^{-1}$ **25/2**

3. Find the value: $\left(\frac{3}{4}\right)^{-1}$ **4/3**

4. Evaluate $2x^2$ when $x = 3$. **18**

Notes

Copy both lines.

I. Substituting: the value goes in parentheses

$$K = \frac{1}{2}mv^2 \quad \text{with} \quad m = 60 \quad \text{and} \quad v = 10:$$

$$\frac{1}{2}(60)(10)^2 = \frac{1}{2}(60)(100) = 3,000 \quad \text{joules}$$

$$\text{with } x = -4: \quad x^2 = (-4)^2 = 16, \quad \text{never} \quad -4^2 = -16$$

Substitute in parentheses, then evaluate by the structure — powers before products.

Notes

Copy both problems.

II. Key words do not decide the operation

15 dollars plus 4 dollars for each of 6 gigabytes:

$$15 + 4(6) = 39$$

15 dollars for each of 4 lines, plus 6 dollars of tax:

$$15(4) + 6 = 66$$

The same words, the same numbers, different structures. Read the situation.

Notes

III. Round once, at the end — and say the unit

$$6\sqrt{200} = 84.8528\dots \approx 84.85 \text{ feet}$$

rounding the side to 14 first: $6(14) = 84 \text{ feet}$

Carry the exact value, round once at the end, and write the unit.

Example 1

A formula with a power in it.

Kinetic energy is $K = \frac{1}{2}mv^2$,

for mass m in kilograms and speed v in meters per second.

Find K for a car of mass 1,200 kg travelling at 15 m/s.

Give the answer with its unit.

Example 1

Worked.

$$K = \frac{1}{2}(1,200)(15)^2 \quad \text{substitute, each value in parentheses}$$

$$= \frac{1}{2}(1,200)(225) \quad \text{the power first — it belongs to the 15 alone}$$

$$= 600(225) = 135,000 \quad \text{then the products}$$

Answer: 135,000 joules

Your Turn

Same steps, your numbers. Boards up when done.

Find $K = \frac{1}{2}mv^2$ for $m = 850$ kg and $v = 12$ m/s.

Your Turn

Answer.

Find $K = \frac{1}{2}mv^2$ for $m = 850$ kg and $v = 12$ m/s.

425 times 144

Answer: 61,200 joules

Example 2

A cost with a rate and a flat fee.

A square floor of side 9 feet is carpeted.

Carpet is 2.89 dollars a square foot and padding is 0.75 dollars a square foot.

Installation is a flat 95 dollars. Find the total cost.

Example 2

Worked.

$$9^2(2.89 + 0.75) + 95 \quad \text{area times the per-square-foot prices, then the flat fee}$$

$$= 81(3.64) + 95 \quad \text{the area and the rate, each finished}$$

$$= 294.84 + 95 = 389.84 \quad \text{multiply, then add}$$

Answer: 389.84 dollars

Your Turn

Same steps, your numbers. Boards up when done.

A square bathroom floor of side 6 feet is tiled.

Tile is 4.25 dollars a square foot and grout is a flat 13.48 dollars.

Find the total cost.

Your Turn

Answer.

A square bathroom floor of side 6 feet is tiled.

Tile is 4.25 dollars a square foot and grout is a flat 13.48 dollars.

Find the total cost.

$36(4.25)$ then add

Answer: 166.48 dollars

Whiteboards • Question 1 of 9

Boards up on three.

Evaluate $3x^2$ when $x = -2$.

Answer it.

Substitute in parentheses.

Whiteboards • Question 1 of 9

Answer.

Evaluate $3x^2$ when $x = -2$.

$$3(-2)^2$$

Answer: 12

Whiteboards • Question 2 of 9

Boards up on three.

A cyclist has a mass of 70 kilograms and rides at 8 meters per second.

Her bicycle cost 480 dollars.

Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v .

Find her kinetic energy.

Answer it.

Whiteboards • Question 2 of 9

Answer.

A cyclist has a mass of 70 kilograms and rides at 8 meters per second.

Her bicycle cost 480 dollars.

Kinetic energy is $K = \frac{1}{2}mv^2$, for a mass m and a speed v .

Find her kinetic energy.

35 times 64

Answer: 2,240 joules

Whiteboards • Question 3 of 9

Boards up on three.

A plan costs 15 dollars plus 4 dollars for each gigabyte.
Find the cost of 6 gigabytes.

Answer it.

What happens six times?

Whiteboards • Question 3 of 9

Answer.

A plan costs 15 dollars plus 4 dollars for each gigabyte.

Find the cost of 6 gigabytes.

$$15 + 4(6)$$

Answer: 39 dollars

Whiteboards • Question 4 of 9

Boards up on three.

A square floor of side 8 feet is tiled at 3.50 dollars a square foot.

Installation is a flat 60 dollars.

Find the total cost.

Answer it.

Which price depends on the area?

Whiteboards • Question 4 of 9

Answer.

A square floor of side 8 feet is tiled at 3.50 dollars a square foot.

Installation is a flat 60 dollars.

Find the total cost.

$$64(3.50) + 60$$

Answer: 284 dollars

Whiteboards • Question 5 of 9

Boards up on three.

Evaluate $\frac{1}{2}mv^2$ when $m = 40$ and $v = 3$.

Which is the value?

A. 180

B. 120

C. 360

D. 3,600

Answer it.

Whiteboards • Question 5 of 9

Answer.

Evaluate $\frac{1}{2}mv^2$ when $m = 40$ and $v = 3$.

Which is the value?

A. 180

B. 120

C. 360

D. 3,600

Answer: A — 180

Whiteboards • Question 6 of 9

Boards up on three.

A square patio of side 12 feet is sealed at 1.25 dollars a square foot.

Delivery of the sealer is a flat 38 dollars.

Which expression gives the total cost?

A. $12(1.25) + 38$

B. $12^2 + 1.25 + 38$

C. $12^2(1.25 + 38)$

D. $12^2(1.25) + 38$

Answer it.

Whiteboards • Question 6 of 9

Answer.

A square patio of side 12 feet is sealed at 1.25 dollars a square foot.

Delivery of the sealer is a flat 38 dollars.

Which expression gives the total cost?

A. $12(1.25) + 38$

B. $12^2 + 1.25 + 38$

C. $12^2(1.25 + 38)$

D. $12^2(1.25) + 38$

Answer: D — $12^2(1.25) + 38$

Whiteboards • Question 7 of 9

Boards up on three.

A fence runs along 6 sides, each $\sqrt{200}$ feet long.

Find its total length, to the nearest hundredth of a foot.

Answer it.

Round once, at the end.

Whiteboards • Question 7 of 9

Answer.

A fence runs along 6 sides, each $\sqrt{200}$ feet long.

Find its total length, to the nearest hundredth of a foot.

$$6 \times 14.1421\dots$$

Answer: 84.85 feet

Whiteboards • Question 8 of 9

Boards up on three.

Evaluate $\frac{a^2 - b}{c}$ when $a = 7$, $b = 4$, $c = 5$.

Answer it.

Finish the top first.

Whiteboards • Question 8 of 9

Answer.

Evaluate $\frac{a^2 - b}{c}$ when $a = 7$, $b = 4$, $c = 5$.

45 over 5

Answer: 9

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

A plan costs 15 dollars plus 4 dollars for each of 6 gigabytes.

Another costs 15 dollars for each of 4 lines, plus 6 dollars of tax.

Find both totals, and say why the key words could not have told you.

Write your answer in sentences.

The same three numbers, the same words.

Whiteboards • Question 9 of 9

Answer.

A plan costs 15 dollars plus 4 dollars for each of 6 gigabytes.

Another costs 15 dollars for each of 4 lines, plus 6 dollars of tax.

Find both totals, and say why the key words could not have told you.

39 and 66

Answer: 39 dollars and 66 dollars

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate variable expressions: word problems — DPG